Cloning-free long-read profiling of somatic hypermutation

Two biological systems test one control-based Oxford Nanopore workflow

1 Complementary test systems

Mouse GC B cells: J_{H4} downstream intron

~1.23-kb amplicon

Wt ($n = 3$)

Aicda$^{-/-}$ ($n = 3$)

Same mice benchmarked by cloned-amplicon Sanger and ONT

N114 hybridoma

Fixed immunoglobulin V region

pMX-*Aicda*

pMX-Empty vector

~0.65-kb amplicon

N114

$n = 3$ independent subclones per condition

2 Shared ONT workflow

R10.4.1 | Kit 14 | SUP basecalling

Intact amplicons are retained as single-molecule reads



substitutions remain phased within each read

Align
target-best competitive alignment

Read QC
length window EE ≤ 0.01/base

Depth
random 500 reads

Call SNVs
confidence + frequency filters

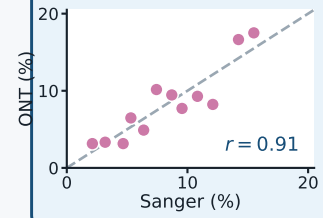
Retain FDR ≤ 0.05 vs. controls and read frequency < 50%
Count each unique position–substitution once per sample

Control-based interpretation

Aicda$^{-/-}$ or empty-vector controls measure residual background

3 Validated outcomes

Sanger concordance



AID dependence

Mouse J_{H4} **≥108×**

N114 V region **≥89×**

G:C-biased AID signature

Longer region

J_{H4}: 1.23-kb intact reads

529-bp Sanger window

1,141-bp reliable analysis

long-range linkage retained

Depth + background

91% J_{H4} **91%** N114

variants recovered at 500 reads

Aicda$^{-/-}$ control background

≈0.55 × 10^{-5} variants/bp

Cloning-free • intact long molecules • depth-standardised, substitution-focused profiling